Where You Are Is What You Get! Inconsistencies of Digital Trace Data Across Download Locations

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ABSTRACT

To collect digital trace data, researchers continue to rely on for-profit companies' Application Programming Interfaces (APIs). These APIs often return samples of the data based on intransparent sampling procedures and algorithms. In this paper, we extend research on the reliability of digital trace data from APIs by examining the effect of the download location on inconsistencies across returned samples: Do we get different values from digital trace data APIs depending on where we download the data from? We compare samples from Google Trends, YouTube Data, and the New York Times (NYT) API from four countries across three continents (Austria, Germany, the U.S., and Australia) for the same query parameters (i.e., search term, region, and time range). Our results show that the download location impacts the returned samples for all three APIs, depending on the query. We find large inconsistencies for samples from Google Trends and the YouTube Data API, while the NYT API returns identical article sets from each download location for most queries. Our findings serve as a cautionary reminder for social scientists relying on sampling-based APIs as they point to yet another limitation regarding their reliability and reproducibility.

Keywords: Reliability, Sample Inconsistencies, Download Location, Google Trends, YouTube, New York Times API

INTRODUCTION

Despite increasingly restrictive access to Application Programming Interfaces (APIs) prompting discussions of a “Post-API age” (e.g., Schoch et al., 2024; Tromble, 2021), APIs often remain the easiest, cheapest, and sometimes the only means of retrieving digital trace data at scale from platforms operated by private, for-profit companies (see, e.g., Ohme et al., 2024). These APIs, however, often return samples of the data based on undisclosed or intransparent sampling procedures and algorithms. Prior research has identified issues with some APIs’ reliability over time and across different API versions. For instance, downloading Google Trends data for identical parameters (i.e., search term, region, time range) at different times can give researchers different values on Google Trends’ search index (Behnen et al., 2020; Eichenauer et al., 2022; Franzén, 2023; Gummer & Oehrlein, 2022, 2024; Mavragani & Ochoa, 2019). Similarly, users and tweets in the returned samples varied depending on Twitter/X’s former API version used to draw the samples (e.g., González-Bailón et al., 2014; Morstatter et al., 2013, 2014; Pfeffer et al., 2018; Tromble et al., 2017). For data collected from the TikTok Research API before July 2024, Pearson et al. (2025) found substantial discrepancies between videos and engagement metrics returned from the API compared to those visible on the website. Such reliability issues present severe challenges for the reproducibility[1] of social science research using APIs based on sampling. For some APIs, such as Google Trends, these reliability issues can be mitigated, though, by averaging the values from trends downloaded on several days (e.g., Gummer & Oehrlein, 2022, 2024; Rovetta, 2021).

In this paper, we expand the research on the reliability of digital trace data from APIs by examining the impact of the download location on inconsistencies across samples. We first examine *whether digital trace data APIs return different values for the same query depending* *on the download country (RQ1)*. Ideally, we would not see any differences so that a researcher downloading data, for instance, in the U.S. should get the same values in the same order for the same query as a researcher in Germany. Second, we test *whether the differences in values across download countries disappear when combining the values from several download days (RQ2)*. With our second research question, we investigate whether researchers can mitigate sampling variance across download countries similarly to inconsistencies across download days by drawing several samples from the APIs or whether differences persist due to relevance- or popularity-based sampling algorithms.

We study these questions by retrieving samples from three popular APIs, namely Google Trends, YouTube Data, and the New York Times' Article Search API (NYT API from here on). They offer free, easily accessible data, making them attractive to researchers. At the same time, the returned values from the APIs are (to varying degrees) based on opaque sampling algorithms, increasing the risk of sampling inconsistencies. However, the way the APIs might work in providing the samples and how transparently they document their functioning differs. Additionally, for Google Trends, previous research has already shown sample reliability issues (Behnen et al., 2020; Eichenauer et al., 2022; Franzén, 2023; Gummer & Oehrlein, 2022, 2024; Mavragani & Ochoa, 2019), while methodological research on their reliability remains scarce for the other two APIs.

We retrieve data from the APIs on seven consecutive days from four countries with varying geographic proximity across three continents (Austria, Germany, the U.S., and Australia). To explore whether query settings impact sample inconsistencies, we vary search query parameters but retrieve samples for the same queries from all four download countries (i.e., queries with the same search term, region, and time range from all download countries). We then compare the samples retrieved from each respective download country, keeping all parameters of the query constant. We preregistered the study details and analysis plan, available at https://osf.io/hdj6q/overview?view_only=05b93d0be2d841cc9d7a8c5345a42579 for Google Trends and at https://osf.io/u742b/overview?view_only=b877a62e5f6b4f82a4d569d7ddb1e1f2 for the YouTube Data and NYT API.

To the best of our knowledge, we are the first to examine the impact of download location on API's sample inconsistencies. We systematically examine the discrepancies in APIs' reliability depending on the download location, while also varying the download day and the query parameters. Doing so, we contribute to a better understanding of how the samples are drawn, where potential discrepancies might come from, and which API specificities might cause them. Our findings and recommendations will help social scientists mitigate the risk of the download location impacting their results and make their research more reproducible in the future.

RELATED WORK

For their research to be reproducible, researchers should get the exact same values from an API every time they download the data from the tool. However, prior research has identified issues with several sampling-based APIs' reliability over time and depending on the API version. Sampling-based APIs are APIs that do not provide researchers with access to the complete data archive (e.g., the Facebook and Instagram API as of April 2026) but only allow them to retrieve samples (i.e., subsets) of the data matching their query (e.g., Google Trends, YouTube Data, and the NYT API). Even if researchers can theoretically download data on the complete archive, rate limits often restrict how many results they can actually retrieve. In cases where the number of results matching a query exceeds the rate limits, researchers can only download a sample of the results, typically in a ranked order, and sampling or ranking algorithms decide which results get returned, often based on opaque criteria. This issue applies, for instance, to the YouTube Data, NYT, Wikipedia, and TikTok API.

In line with the large body of social science studies using Twitter data, Twitter/X's former APIs are the most well-researched in this regard (e.g., González-Bailón et al., 2014; Morstatter et al., 2013, 2014; Pfeffer et al., 2018, 2023; Tromble et al., 2017). Studies usually compared the values returned from Twitter's free, sampling-based APIs, such as the Sampling, Streaming, and Search API with values from the full data archive, for instance, from the paid Firehose access. While some studies found only few systematic discrepancies (e.g., Morstatter et al., 2014), other researchers detected substantial inconsistencies (González-Bailón et al., 2014; Morstatter et al., 2013; Pfeffer et al., 2018; Tromble et al., 2017). The differences were larger when the researchers queried for fewer search terms matching the tweets' content and for smaller samples (González-Bailón et al., 2014; Morstatter et al., 2013). However, X's latest API release allows for paid access to the full data archive, largely reducing the reliability issues coming from sampling (Pfeffer et al., 2023).

Researchers have also extensively studied Google Trends. Several studies have demonstrated that downloading Google Trends data for the same parameters (i.e., search term, region, time range) but at different points in time produces different values on Google Trends' search index (Behnen et al., 2020; Eichenauer et al., 2022; Franzén, 2023; Gummer & Oehrlein, 2022, 2024; Mavragani & Ochoa, 2019). The severity of the sample instability depends on the query's search volume. While samples for high search volume queries tend to be consistent, inconsistencies are substantial for low search volume queries, such as less popular search terms, shorter search time ranges, and smaller regional units, including searches from smaller countries and countries with low internet penetration rates.

Most recently, researchers noticed inconsistencies in the data returned from the TikTok Research API. For data collected before July 2024, Pearson et al. (2025) find substantial discrepancies between videos and engagement metrics returned from the API compared to those visible on the website. The specific issue seems to have been resolved in July 2024, as the researchers only found marginal differences after that (Pearson et al., 2025). In June 2025, however, new reliability concerns emerged about the TikTok API not providing metadata for many videos (AI Forensics, 2025).

For all these APIs, researchers have long criticized the lacking documentation, intransparent sampling procedures and algorithms, and the APIs overall being a black box that need continuous academic monitoring (e.g., Hölzl et al., 2025; Morstatter et al., 2014; Pearson et al., 2025; Pfeffer et al., 2018). Our study builds on this work and is the first to examine APIs' reliability issues not only depending on the download day and API version, but the download location and the interlink between download day and location.

USE CASE: THE GOOGLE TRENDS, YOUTUBE DATA, AND NEW YORK TIMES' ARTICLE SEARCH API

To answer our research questions, we select three popular sampling-based APIs as use cases, namely Google Trends, YouTube Data, and the New York Times' Article Search API[2], with 88,970, 2,334, and 2,174 research article hits, respectively, in the ScienceDirect database at the time of writing (ScienceDirect, 2025a, 2025b, 2025c). Google Trends[3] offers aggregated information on the relative frequency of Google searches for specific terms by region and over time, reported as a normalized Relative Search Index (RSI) ranging from 0 to 100 (see, e.g., Google, 2026; Stephens-Davidowitz & Varian, 2015). Mostly in the form of time series, this type of Google search data is increasingly used in social science studies (e.g., Hölzl et al., 2025). A large body of research has also made use of the YouTube Data API that provides metadata, including views, likes, and comments for videos uploaded on the world's most popular video platform (Auxier & Anderson, 2021; Breuer et al., 2023; Konijn et al., 2013; similarweb, 2025). Researchers using its search function receive sets of videos matching their query (i.e., a specific search term, video upload region, and upload date). The New York Times (NYT) has provided a robust API for research since as early as 2009 and has since expanded its offerings to include several APIs with immense research potential, such as the Archive API, the Books API, and the Article Search API (Gottfrid, 2009; Jacobson et al., 2012; The New York Times Developer Network, 2026; Thorp, 2010). With the Article Search API (NYT API), users can download sets of articles matching their query (i.e., search term, publication region, and publication date) from an extensive collection of articles published in the NYT (The New York Times Developer Network, 2026).

We selected these APIs as they have some common features but also some differences in the way they function. They all offer free, easily accessible data, making them attractive to researchers. At the same time, the returned values from the APIs are (to varying degrees) based on opaque sampling algorithms, increasing the risk of sample inconsistencies. However, the way the APIs might work in providing the samples (i.e., random sampling versus rank-based sampling algorithms), how transparently they document their functioning, and the volume of the data they process (i.e., number of searches, videos, and articles) differs. Additionally, for Google Trends, previous research has already shown reliability issues (Behnen et al., 2020; Eichenauer et al., 2022; Franzén, 2023; Gummer & Oehrlein, 2022, 2024; Mavragani & Ochoa, 2019), while methodological research on their reliability remains scarce for the other two APIs.

Google Trends and the YouTube Data API are based on proprietary architectures. Although researchers can input queries and receive results from these APIs, they have no insight into the internal mechanisms that generate those results. We thus have to speculate on the underlying architectures based on what is known about the APIs' behaviors and the common data analytics architectures in the industry. From this speculation, we derive our expectations for the temporal and geographic inconsistency in query results. In contrast, the NYT API is a lot more transparent, and based on the available information, we can make assumptions on the likely outcome for geographic and temporal variation in query results. In the following, we elaborate on our educated assumptions and expectations for each API's architecture in more detail.

Assumptions on Google Trends' Architecture

According to Google (Google, 2026), Google Trends' RSI is derived from a random sample of all Google searches. However, Google does not disclose details of how, when, and where they draw these samples. Google Trends is likely built on a type of data warehouse architecture or a similar big data storage technology with many regional replicas of the data. We can presume these architecture decisions are the case because, without regional replicas, service time for users would be undesirably slow (Mokadem & Hameurlain, 2020). When requesting to download data from Google Trends, the API may redirect the user to a replica geographically close to the download location. The data may be separately updated over time and asynchronously harmonized across regional replicas, which might in turn lead to inconsistent samples over time and across download locations (see, e.g., Inmon, 2005;

Kemme & Alonso, 2000; Sullivan, 2020). Google Trends' objective is to provide an account of relative search term frequency across time periods and regions, for which the "true" frequency for a given term is fixed once that period has ended. Query results from a given regional server may thus change as new data becomes available during data store synchronization, especially when requesting more recent data and for queries matching smaller samples.

Assumptions on the YouTube Data API's Architecture

In contrast to Google Trends, researchers can theoretically download the full data archive from the YouTube Data API. The number of videos matching a query, however, can easily exceed the API's rate limits (maximum of around 2,310 videos per day). In these cases, the YouTube Data API only returns a sample (i.e., subset) of the data with videos ranked in a specific order. The YouTube Data API is likely also built on a different architecture than Google Trends, as the API's goal is to provide live data on videos rather than retrospective information, as is the case with Google Trends. The YouTube Data API is known to update in real-time based on current events such as shifts in video popularity and changes in a user's search history (Ng et al., 2023). Given that YouTube is also managed by Google, which develops its own big data analytics solutions, it is likely that the platforms supporting the YouTube Data API may be a system like BigTable (Chang et al., 2008) or Spanner (Corbett et al., 2013). Both systems ingest large amounts of data and replicate it across geographies with periodic synchronization at varying velocities. Consequently, as a user's watch history accumulates and video trends evolve, the output of the YouTube Data API might vary, in addition to the periodic synchronization of geographic replicas. Similarly to YouTube's complex recommendation system (e.g., Covington et al., 2016), result sorting patterns by default follow a relevance-based sampling approach that may vary by day and location (Google for Developers, 2026).

Assumptions on the New York Times Article Search API's Architecture

Much like the YouTube Data API, the NYT API's rate limits can prevent researchers from downloading all articles matching their query (maximum of 5,000 articles per day), limiting them to retrieving only a sample of the data. Compared to Google Trends and YouTube, though, the NYT API is notably more transparent about its underlying architecture. If users request less than the total number of articles matching a query, the NYT API can return values derived from a relevance-based algorithm. Even though no precise information on how the relevance-based algorithm samples articles is available, the NYT at least provides detailed documentation for developers, including openly published code (Mandeville et al., n.d.; The New York Times Developer Network, 2026). This documentation reveals that the API uses Elasticsearch (Gormley & Tong, 2015) to rank a large corpus of documents based on the relevance of their text to a query. Elasticsearch operates as a document store, which differs from data warehouses and other storage systems previously discussed. A document store is optimized for rapid full-text search across a large corpus of text rather than for high-speed operations like record updates. Regarding consistency in query results over time and across geographies, the NYT's document store is likely replicated in multiple locations to protect against losing data from server failures and to provide timely global service (Mokadem & Hameurlain, 2020). However, this replication may introduce some synchronization delays depending on the user's location. Given that the data primarily consist of news articles, a smaller volume of data to be synchronized compared to Google searches or YouTube videos, we expect the NYT API to exhibit less variation in query results over time and across different regions.

In summary, while Google Trends' RSI is supposedly derived from a random sample of all Google searches, the YouTube Data and NYT API might rely on relevance- and popularity-based sampling, potentially varying by day and location. These APIs' sampling algorithms might thus return videos and articles deemed more relevant for a given download day and country. The systems supporting Google Trends and YouTube remain opaque, while we have a clearer understanding of the architecture underlying the NYT API. Because YouTube Data API results vary in real-time based on factors such as video popularity, it likely employs a different system from Google Trends, although both may draw on Google's big data analytics platforms. In contrast, the NYT API's document store synchronizing an overall lower volume of news articles across geographic replicas should yield more consistent results across time and place.

METHODS

To answer RQ1, namely whether values returned from the APIs for the same query on the same day differ depending on the download country, we first examine API reliability issues stemming from the download location. To study the relationship between download day and download location and to identify potential mitigation strategies, RQ2 investigates whether the differences in values across download countries disappear when combining the values from several download days. In the following, we describe our data collection, analysis approach, and expectations for our research questions.

Data Collection

Following our preregistered analysis plan, we collected data from the APIs[4] on seven consecutive days (April 18 to 24, 2024, from Google Trends and August 11 to 17, 2025, from

YouTube and the NYT API[5]) from four countries (Germany, Austria, the U.S., and Australia). We selected these four download countries as use cases as they vary in country size and geographic proximity, and span three continents. At the same time, they share some cultural affinity (especially Germany and Austria), mitigating the impact of cultural proximity on potential inconsistencies across download countries. We simulated downloading the data from each country by using a VPN connection from every country's biggest city (Berlin, Vienna, New York, and Sydney). To mitigate any temporal variance between the downloads from different countries within the same day, we executed the downloads immediately one after the other. We collected the data using R version 4.4.1 (2024-06-14) and the R packages \`gtrendsR\` (Massicotte & Eddelbuettel, 2022) for Google Trends, \`tuber\` (Sood et al., 2022) for YouTube Data, and \`jsonlite\` (Ooms, 2014) for the NYT API.

For all three APIs, researchers can vary the query's search term(s), time range, and region, namely when and where the searches for the term have been conducted on Google, when and where the videos have been uploaded on YouTube, and when and where the articles have been published in the New York Times. Following Behnen et al. (2020), we systematically varied these search query parameters but retrieved the same queries from all four download countries (i.e., queries for the same search term in the same language, region, and time range from all download countries). We selected the query parameters in an exploratory manner based on previous research (Behnen et al., 2020). Specifically, we modified the queries in the following way:

- Three search terms in either German (Google Trends and YouTube) or English (NYT) with high, medium, and low search, video, and article volume. For Google Trends, we again followed Behnen et al. (2020)'s parameter selection. For YouTube Data and the

NYT API, we determined the videos' and articles' upload volume in an exploratory way by looking at the APIs' automatically returned number of total results.

- Four time ranges spanning over ten years to one week.
- Two regions including worldwide searches, videos, and articles, as well as searches, videos, and articles from Germany.

Using these queries allows us to descriptively explore how the target population size of searches, videos, and articles might impact download inconsistencies. In total, we downloaded samples from each country and on each day for 24 parameter combinations (see Table 1 for a complete overview of our data collection approach and query parameters, and our preregistration for more detailed reasoning behind our data collection approach[6]). For each combination, we retrieved all available returned values from Google Trends and, at maximum, the first 50 results from the YouTube Data and NYT API to not exceed the rate limits. In case of errors from the APIs, we retried querying the API a second time. For some queries, however, the APIs returned fewer than 50 results or even zero when no results matched the specific combination of search term, time range, and region (e.g., no videos have been uploaded to YouTube in the last week in Germany containing the term 'banana milk'). Tables A1 to A3 in the Appendix provide an excerpt of what the samples look like after being downloaded from each of the APIs.

[insert Table 1.]

Analyses

After the data collection, we compare the samples retrieved from each respective download country, keeping all parameters of the query constant. Google Trends returns a time series for each query as the normalized RSI, representing the relative amount of searches for the query over time with values ranging from 0 to 100. In contrast, YouTube and the NYT API provide lists of videos and articles in a specific order. We thus tailor our analyses to the specificities of the different APIs instead of conducting the same analyses for all three of them. For YouTube and the NYT API, we not only analyze the sets of videos and articles we retrieve but also the order in which they are returned. Doing so, we can draw conclusions on whether researchers might get different results if they only retrieve the, for instance, first ten compared to the first twenty videos and articles.

For Google Trends, we first replicate the analyses conducted by Behnen et al. (2020) and, second, add analyses to examine the time series' similarities. Our first research question asks whether APIs return different values for the same query depending on the download country. To answer RQ1, we (1) calculate the absolute and relative standard deviation of the RSI values across download countries for each download day for each download country. We (2) run Mann-Whitney-U-tests for pairwise country-day-combinations. Expanding the analyses conducted by Behnen et al. (2020), we (3) calculate the Euclidean distance standardized by the time series length for the RSI values for each download country pair[7] and (4) regress the Euclidean distance on the download country pair, the download day, and the query parameters to identify the main factors behind any observed inconsistencies[8]. If we do find differences across download countries, we go on to examine RQ2, whether the differences in the samples across download locations disappear when combining samples from several download days. We run the same analyses as for RQ1 but use the median RSI values of the seven download days instead of the values from each individual day[9].

For YouTube Data and the NYT API, we explore the samples' similarity by comparing the returned articles' and videos' IDs and their ranks. We examine whether we retrieve the same videos in the same order from each download country on each day, keeping all query parameters constant. To investigate RQ1, we (1) calculate the videos' and articles' set-based Jaccard distance and regress the distance values on the download country pair, download day, and query parameters. Additionally taking the article and video order into account, we (2) calculate the edit-based Levenshtein distance and the Rank-Biased Overlap (RBO) distance, and again regress their values on the download country pair, download day, and query parameters. Lastly, we (3) examine the videos' and articles' ranks' absolute and relative standard deviation and Pearson correlation coefficients. If we do find differences across download countries, we again go on to examine RQ2. We adapt the analyses from RQ1 by pooling the sets of video and article IDs from each day for each query and download country as their union (i.e., video and article IDs that are in the sample from any day), and calculate the Jaccard distance of the union sets for each download country pair. To take the article and video order into account for RQ2, we calculate the absolute and relative standard deviation and correlation coefficients across download countries for the median rank of all download days. As they are not set-based metrics, we refrain from calculating the Levenshtein and RBO distance for RQ2.

Expectations

Assuming that APIs rely on asynchronously updating regional data replicas, we expect differences in the APIs' returned values for the same query on the same day downloaded from different countries (RQ1). If replicas synchronize over time, we expect pooled samples from several days to be more similar than the samples from the single days (RQ2). However, if the APIs' sampling algorithms take download location into account for the returned values, pooled samples may not reduce cross-country inconsistencies.

While examining RQ1 and RQ2, we take a more detailed descriptive look at differences across (1) pairs of download countries; (2) download locations depending on the query parameters; and (3) download days within the same download country:

(1) We explore whether discrepancies are more pronounced when comparing some download country pairs to others. For instance, we might not find differences between neighboring countries (e.g., Germany and Austria) but might find differences between more geographically distant countries from different continents (e.g., Germany and Australia). We expect the geographically more distant countries to have a significantly larger impact on the distance between Google Trends time series and video and article sets than when comparing Germany and Austria.

(2) For Google Trends, prior findings indicate that a smaller overall search volume leads to greater sample inconsistencies (Behnen et al., 2020; Eichenauer et al., 2022; Gummer & Oehrlein, 2024; Mavragani & Ochoa, 2019). For instance, RSI values showing the less frequent searches for 'short-time work' in Germany during the last week might present more variation across download locations and days than the RSI values showing worldwide searches for 'sofa' from 2010 to 2023. Similarly, queries with a lower number of matching videos and articles may exhibit greater inconsistencies. We will therefore examine whether differences in returned RSI values, videos, and articles across download countries are (significantly) larger for queries with terms with lower search, video, and article volumes, for smaller time ranges, and for smaller regions.

(3) We also examine whether discrepancies across download days differ by download country. Differences in RSI values, videos, and articles across download days might, for instance, be (significantly) smaller for bigger download countries such as the United States and (significantly) larger for smaller download countries such as Austria.

For more details on our data collection, preparation, and analysis, see our preregistrations and open source data and code.

RESULTS

Our findings show inconsistencies across download countries and also download days for all three APIs, albeit to very different extents. In this section, we describe the results for the Google Trends, YouTube Data, and NYT API in relation to each research question. We mainly focus on describing the findings based on the Euclidean distance for Google Trends and the Jaccard distance for the YouTube Data and NYT API as all metrics show a very similar picture. Adding to our preregistered analyses, we also conducted several robustness checks that we had not preregistered. A detailed description of the results from the other metrics and additional analyses can be found in Online Appendix OA to OD.

Results for Google Trends

Answering RQ1a: Does Google Trends return different values for the same query depending on the download country?

Consistent with previous research on reliability issues stemming from the download day (e.g., Behnen et al., 2020; Eichenauer et al., 2022; Franzén, 2023; Gummer & Oehrlein, 2022, 2024; Mavragani & Ochoa, 2019), we find that Google Trends returns inconsistent samples across download countries for several queries. The extent of the inconsistencies depends on the download day, the overall search volume and thus population size, and the download countries' geographic distance. Based on the Euclidean distance, Figure 1 shows the differences in the time series for each download country pair for each day in the columns and for each query in the rows. Higher Euclidean distance values imply that the time series from different download countries are more different (marked in darker shades of red).

[insert Figure 1.]

With an Euclidean distance of zero, the samples downloaded from Germany and Austria are identical or almost identical for all queries. In contrast, there are differences in the time series when comparing all other download countries with one another. The trends downloaded from Australia differ the most from the trends downloaded from the other countries (e.g., maximum Euclidean distance between the trends from Australia and Germany downloaded on day 2 for 'short-time work' searched for in the last seven days in Germany). The more distant the download countries are geographically, the more different the time series Google Trends returns.

Turning to the findings on the queries' overall search volume, high-volume queries matching the search term 'sofa' show little to no variation across download locations (average Euclidean distance of 0.2, ranging from 0 to 0.5; see Table A4 in the Appendix). In contrast, inconsistencies across download countries are most pronounced for the low (average across all 'short-time work' queries of 0.4, ranging from 0 to 3.4), followed by the medium search volume term (average across all 'roofer' queries of 0.2, ranging from 0 to 1.1), for data from the last seven days (average of 0.5), and for searches in the smaller geographic region of

Germany compared to worldwide searches (average of 0.3 versus 0.2). Figure 2 shows what the time series from the different download countries look like for the lowest search volume query with the highest Euclidean distance. In case of consistent samples, the trends from the different download countries should perfectly overlap, which would imply that the same values are returned from Google Trends from each download country. While the trends from Germany and Austria mostly overlap, the trends from the U.S. and Australia look similar but are not identical. Not only do the trends not overlap with the trends from the other download countries, the maximum value of 100 to which all other values of the index are relative is also on different days. These discrepancies in the maximum lead to the shifts in the trends from the different download countries.

[insert Figure 2.]

Figure 3 shows the results of regressing the standardized Euclidean distance between two download countries on the download country pair, the download day, and the query parameters. With the exception of queries for the last five years compared to 2010 to 2023, all descriptively observed effects are statistically significant at the 0.05 level. Looking at the main factors behind the observed inconsistencies, we see the largest effect size for data downloaded for the last week (again relative to 2010–2023).

[insert Figure 3.]

Answering RQ2a: Do the differences in samples from Google Trends across download locations disappear when combining the samples from several download days?

Existing differences can be reduced when using the median of RSI values downloaded on different days instead of the trends from a single day. For medium to high search volume queries that already start with relatively consistent samples, this approach leads to almost identical trends from the different download countries. For instance, the Euclidean distance for the trends for the term 'roofer' searched for in Germany between 2010 and 2023 from the U.S. compared to Germany decreases from around 0.3 to 0.1 (see Table A5 in the Appendix). Figure 4 shows to what extent these median trends overlap across download countries. However, for low search volume queries such as searches for the terms 'short-time work', using the median only reduces the Euclidean distance between the trends from Australia and Germany from 3.4 to 1.1. We see a similar decrease but no erasure of discrepancies for these queries for the other download country comparisons as well. Using the median can thus account for the sampling variance for high search volume queries, but does not lead to consistent samples for low search volume queries.

[insert Figure 4.]

The results from the Pearson correlation coefficients, Mann-Whitney-U-tests, and absolute and relative standard deviations mirror our findings from the Euclidean distance. Noticeably, though, the time series from all download countries strongly and significantly correlate with each other. We also see a similar picture when looking at how much the samples downloaded from the same country but on different days vary, with the lowest search queries showing the largest inconsistencies. The inconsistencies do not seem to systematically depend on the specific download country or the download country size (for details, see Online Appendix OA).

Results for the YouTube Data API

Answering RQ1b: Does the YouTube Data API return different values for the same query depending on the download country?

The YouTube Data API exhibits by far the greatest reliability issues of the APIs under consideration, with returned videos differing largely depending on both the download country and the download day. Figure 5 shows the set-based Jaccard distance for videos returned from the YouTube Data API for each query, comparing the samples from each download country with one another on each download day separately, and for the union of videos returned on any of the days. The Jaccard distance can range from 0 to 1, with higher values displayed in a darker shade of red. For most country comparisons, the Jaccard distance indicates large discrepancies across download countries up to the maximum distance of 1. We find this maximum value for samples downloaded on day 4 or 6 from Germany compared to Australia and the U.S., Australia compared to the U.S., and Austria compared to the U.S. for videos on “Fitness” uploaded in Germany within the last week. As the Jaccard distance ranges from 0 to 1, this value shows that none of the 50 videos returned for the same query from one download country is also included in the sample from the other download country.

[insert Figure 5.]

Once again, we see the least differences for the two neighboring download countries, Germany and Austria. Contrary to Google Trends, though, we also see fewer differences when comparing the samples from the U.S. and Australia. For low video volume queries, the Jaccard distance is even zero for these download country comparisons, meaning that identical samples have been returned from the two download countries.

Overall, we see the largest differences for the queries with the largest video volume, so for videos matching the most popular search term “Fitness” (Jaccard distance between 0 and 1 with average of 0.64; see Table A6). Smaller differences occur for videos on the medium video volume term “short-time work” (Jaccard distance between 0 and 0.73 with average of 0.21) and the low video volume term “banana milk” (Jaccard distance between 0 and 0.82 with average of 0.26). When only a few videos match a query, the API thus returns mostly identical videos from every download country. We see a similar picture when comparing the samples from the same download country but on different download days.

Not only do we not get identical video sets from different download locations for high video volume queries. The videos that do overlap in the sets from different download countries are also not returned in the same order as the Levenshtein distance, distance from RBO, the video ranks’ absolute and relative standard deviations, and their correlations show. We see high discrepancies in the videos’ order for all download country comparisons with high video volume queries again displaying the largest discrepancies (see Online Appendix OB). Figure 6 reports the results of regressing the Jaccard distance between download country pairs on the download and query parameters. The estimates indicate that download country pairs and most query parameters again have a statistically significant effect on the distance between video lists retrieved from different countries. Where the data has been downloaded from (i.e., the download country pairs) shows the largest impact on the inconsistencies. In contrast, the download day does not appear to influence inconsistencies across download locations.

[insert Figure 6.]

Answering RQ2b: Do the differences in samples from the YouTube Data API across download locations disappear when combining the samples from several download days?

In contrast to Google Trends, downloading samples on several days and pooling them does not help in reducing the sample inconsistencies across download countries for the YouTube Data API. As an illustration, we take a look at the query with the largest discrepancies across download countries. Using the union of video sets of the seven download days reduces the Jaccard distance from the maximum value of 1 for download day 6 to 0.68 for samples from the U.S. compared to Austria (see Table A7). This value implies that only one-third of the videos are identical in the sets from both download countries. As a comparison, we look at a low inconsistency query, such as videos on 'short-time work' uploaded in Germany in the last week. While the samples from the U.S. and Austria downloaded on day 5 show a distance of 0.5, the Jaccard distance for the union still lies at 0.25. Using the union of videos returned on any download day does thus not provide a sufficient strategy for researchers to eliminate the YouTube Data API's reliability issues. When looking at the order in which the videos have been returned on different download days, however, the videos' order becomes substantially more consistent when using the median rank (see Online Appendix OB).

Results for the NYT API

Answering RQ1c: Does the NYT API return different values for the same query depending on the download country?

Compared to Google Trends and the YouTube Data API, the samples from the NYT API show the least inconsistencies. Except for the queries with the lowest article volume, the API returned the same articles in almost the exact same order from each download country on each day. Figure 7 again shows the Jaccard distance for each query, comparing the samples from each download country on each download day separately, and for the union of articles returned on any of the days. Higher values are displayed in a darker shade of red. In our comparison of samples from the NYT API from different download countries, the Jaccard distance ranges from 0 for the large majority of queries to a maximum of 0.41. We find this greatest inconsistency for the comparison of samples from Austria and the U.S. downloaded on day 2, including articles on the search term ‘sofa’ published worldwide in the last month. For this case, 17 of the 27 returned articles are returned in the samples from both download countries. Overall, only queries for the search term ‘sofa’ published worldwide in the last five years, last month, or last week show any discrepancies across download countries with a Jaccard distance greater than zero (average distance across all ‘sofa’ queries of 0.08, see Table A8). The differences are similar for all download country comparisons (e.g., when comparing Germany to Austria versus when comparing Germany to the U.S.), and we see some seemingly random variance in the discrepancies’ size across download days (e.g., on some days some download country comparisons are larger than others, while on other days we see no difference).

[insert Figure 7.]

When looking at the Levenshtein and RBO distance and the article ranks’ absolute and relative standard deviation and correlations, we again find inconsistencies in the article order for the search term matching the least articles. However, the article order also varies to a very small extent for the two more popular search terms (e.g., two article pairs in switched order). We again see a similar picture when comparing the samples from the same download country but on different download days, for both the article sets and the article order. The inconsistencies do not seem to systematically depend on the specific download country or the download country size (for details, see Online Appendix OC). This finding is also reflected in the insignificant download country pair effects when regressing the Jaccard distance between download country pairs on the download and query parameters (see Figure 8). In contrast, the search term's article volume appears to be the largest impact factor.

[insert Figure 8.]

Answering RQ2c: Do the differences in samples from the NYT API across download locations disappear when combining the samples from several download days?

Existing differences in the article sets and order across download locations largely decrease or disappear completely when using the union of articles from all days compared to the single days (see Table A9). For instance, articles on the search term 'sofa' published worldwide in the last month initially show the largest inconsistencies across download countries. For this query, the Jaccard distance decreases from 0.41 on download day 2 to 0.13 for the union of articles, with 28 of 30 articles being shared in the union samples from the U.S. and Austria. The article order also becomes more consistent when using the ranks' median, but existing differences do not completely disappear (see Online Appendix OC).

DISCUSSION

Our results show that both download location and download day impact the returned samples for all three APIs, albeit to very different degrees. The extent of inconsistencies largely depends on the API and the query. The NYT API yields the most consistent samples, returning identical article sets for almost all queries. In comparison, Google Trends and the YouTube Data API show larger discrepancies, with YouTube Data returning the least consistent samples.

This finding is in line with the fact that the NYT API is not only the most transparent in how it is built but also processes the smallest data volume. The NYT API returns identical article sets for larger article volume queries (e.g., on 'politics'), while for less common search terms, pooling samples from several download days can mitigate the impact of the download country.

Consistent with previous research (Behnen et al., 2020; Eichenauer et al., 2022; Franzén, 2023; Gummer & Oehrlein, 2022, 2024; Mavragani & Ochoa, 2019), Google Trends exhibits larger inconsistencies across download countries than the NYT API. We again see significantly more inconsistencies across download countries, the lower the query's search volume (i.e., lower search volume terms, shorter time span, and searched for in Germany instead of worldwide searches), and when comparing more geographically distant download countries. We find smaller or even no differences for queries including the high search volume term and when comparing samples from the two neighboring European countries, Germany and Austria. For high-volume queries, using the median across download days reduces sampling variance.

Our findings support the assumption that the NYT API and Google Trends use regional data replicas that asynchronously synchronize over time. Sampling variance is larger for low search and article volume queries, where the samples come from a smaller population of searches and articles. Pooling samples from several download days can reduce the sampling variance for both APIs. However, the results derived from Google Trends for searches from the last week still depend on the download location, even after using the samples' median from several days. The regional replicas might thus not synchronize quickly enough to counter the larger sampling variance of this low search volume query. The overall data volume (all Google searches versus all NYT articles) might further impact how quickly the data replicas can synchronize, with larger data size leading to longer synchronization times.

The YouTube Data API shows by far the largest inconsistencies across both download countries and days, especially for queries matching a large number of videos. For these popular queries, the sampling algorithm has more videos to choose from and apparently takes download location and download day into account when returning the samples. When only a few videos match the query, researchers get consistent samples across download locations and download days. We also see fewer inconsistencies for samples downloaded from Germany compared to Austria and from the U.S. compared to Australia. This finding suggests that cultural similarities across download locations might play a larger role for the algorithm in what videos get returned than the geographic distance of the countries. Contrary to the NYT API and Google Trends, downloading samples on several days and taking the union of video sets does not sufficiently reduce these reliability issues.

For the YouTube Data API, inconsistencies seem to not only stem from asynchronous data replicas. Instead, these findings speak to the assumption that the algorithm returns videos that it deems most relevant to the user's download location and download day[9]. In some cases, researchers might even want to exploit the geographical variation by examining what users from different countries get to see for the same search term. However, even for this type of research, the reproducibility issues remain, as researchers do not have insight into the APIs' precise functioning and into how these differences come about. Regardless of whether inconsistencies across download locations arise from random sampling variation or from location-dependent, rank-based sampling, researchers face the same reliability and reproducibility issue. They may obtain different data from the APIs for the same search term, language, region, and time range, depending on the download country and day.

CONCLUSION

For research to be reproducible, a researcher downloading data from an API at one point in time from one country needs to get identical data as a researcher downloading data using the exact same query at a different point in time from a different country. Extending previous research showing that the samples returned from an API can depend on the download day, we examine whether the returned samples also depend on the download location. We downloaded data from the Google Trends, YouTube Data, and NYT API from four different countries on seven days for identical queries and systematically varied the query parameters. Our results show that data collection from sampling-based APIs is impacted by the download country, pointing to an additional limitation to the reliability and reproducibility of API-based social science research. The reliability issues' sources, their extent, and how they can be mitigated, however, largely depend on the specific API used. While inconsistencies across download locations from Google Trends and the NYT API appear to stem from asynchronously updating data replicas, sampling algorithms seem to play a role for the YouTube Data API.

Our results lead to practical recommendations well beyond the three example APIs. When retrieving data from any sampling-based API, we strongly recommend researchers to check whether they get different results from the API for an identical request depending on their download country, for example, by using a VPN connection. If the download location does impact the returned samples, researchers need to carefully consider the API's specificities. They then must either take additional steps to mitigate this impact or seriously reconsider collecting the data through the API if the returned data are heavily impacted by the download location and the API's sampling process remains a black box. As a minimum requirement, researchers should state the date and location of data retrieval and critically discuss potential reliability issues of their data. For some APIs (e.g., Google Trends, the NYT API, and similar sampling-based APIs), researchers can mitigate sample inconsistencies across download locations and also download days by using the median or union of samples from several download days from a single download country. Depending on the specific API, researchers should consider if their construct of interest allows the retrieval of data on high (APIs such as the NYT API or Google Trends) or low (relevance-based APIs such as the YouTube Data API) search volume queries as these might yield more consistent samples. Whenever possible, researchers should download all results matching a query to avoid getting a sample or subset of the data and choose sampling-based APIs that are transparent in their functioning. By adopting these practices, researchers can enhance the reproducibility and robustness of their findings when using digital trace data from APIs. If researchers want to avoid these platform-related issues altogether, they might want to consider user-centered forms of digital trace data collection, such as data donation (see, e.g., Ohme et al., 2024).

While clearly showing that download location can impact returned samples from APIs, our research also comes with several limitations. The first limitation is inherent to any research using data from proprietary APIs from for-profit companies. As these companies do not tell researchers how they precisely built their APIs, we can only speculate about the mechanisms that lead to these sample inconsistencies. For Google Trends, for instance, Google draws a new, supposedly random sample of all Google searches each day (Google, 2026). We cannot know, however, whether the sampling happens at the same time in each data warehouse or depends on the regional replica's time zone. Time zones might thus impact the observed inconsistencies across download countries. Additionally, the APIs' functioning can change at any time or the APIs can even cease to exist. For instance, we do not know whether Google will discontinue the Google Trends website after the introduction of the new Google Trends API (Google Search Central, 2025). Future research might thus assess to what extent download location and download day still impact samples from the new Google Trends version. Second, we only provide empirical evidence for the download location impacting returned samples for the three APIs under consideration. Previous research, for instance, on the TikTok API's reliability issues (e.g., Pearson et al., 2025), suggests that our findings might very well translate to other sampling-based APIs as well. Finally, our comparison of download countries is restricted to four different countries. We selected these countries due to their geographic and cultural proximity and distance. Whether the countries' geographic or cultural distance leads to the observed inconsistencies, however, might depend on each API's functioning. To further assess whether cultural rather than geographic proximity influences returned values, for instance in the case of YouTube videos, incorporating a more diverse and extensive set of download countries would be valuable.

In sum, our study highlights the importance of download location as a potential source of inconsistencies in digital trace data from sampling-based APIs. Our findings point to another limitation when relying on APIs for social science research, especially regarding the reliability and reproducibility, but also the potential generalizability of findings derived from API-based data. They serve as a cautionary reminder for computational social scientists and call attention to the need for greater transparency from API providers regarding their sampling procedures.

NOTES

1. Following Schoch et al. (2024), we use the term *(computational) reproducibility* to refer to re-running a study's code to obtain the same numerical results. In computational social science, this process often includes re-collecting data if data acquisition, for instance, via API calls, is part of the code. While *reproducibility* traditionally involves re-running analyses with the original data, and *replication* refers to repeating a study with new data, API-based research blurs this distinction.
2. With its increasing popularity in social science research and known reliability issues, the TikTok API might have been another well-suited use case. Access to the API is currently restricted to researchers from the U.S. and Europe, however (TikTok for developers, 2025). At the time of preregistration, the API had additionally only granted access for research on extremism and related topics in line with the EU's Digital Services Act (de Bailliencourt, 2022; Kupferschmidt, 2025), leading us to refrain from using the TikTok API as one of our exemplary APIs.
3. Strictly speaking, the current version of Google Trends itself is not an API but a website providing data. R packages and Python modules such as gtrendsR (Massicotte & Eddelbuettel, 2022) and pytrends (DeWilde & Hogue, 2022) are taking on the function of an API by scraping the website and functioning as the intersection between the website and the user. In July 2025, Google announced a new Google Trends API (Google Search Central, 2025). According to Google, this API should tackle some previous usage issues of Google Trends for research. We applied for early access to the API's alpha version in July 2025 but have not heard back from Google at the time of writing. The new Google Trends API also is not currently available to researchers at large, leaving the initial Google Trends version as the main access point for data collection.

4. We successfully applied to the YouTube Researcher Program (YouTube n.d.) and received approval for our research project. We queried the YouTube Data API from a new gmail-account that has not previously been connected to YouTube and disabled the tracking of the account's YouTube watch history. For the main preregistered analysis, we obtained OAuth authentication. Google Trends and the NYT APIs do not require project applications before collecting data from the APIs. For the NYT API, we obtained an API key upon registration, while Google Trends is accessible without registering.

5. To conduct several robustness checks, we repeated the data collection process over seven days for YouTube Data and the NYT API, in April 2025, August 2025, and March 2026. In the main text, we report our findings from August 2025 for both APIs as this data collection showed substantially larger discrepancies across download countries for the NYT data compared to the first round of data collection. For the YouTube Data API, we see almost no differences across the three data collections. The results from our robustness checks can be found in detail in Online Appendix OD.

6. The NYT API was updated between our preregistration and data collection period. The update changed how some parameters are called, leading to minor changes in our code compared to our preregistration. For the YouTube Data and NYT API, researchers can set a parameter defining by which criterion the API should sort the returned results. For the main analysis, we set this parameter to 'relevance' but conducted robustness checks setting the parameter to 'date' and 'rating' (YouTube) and 'date' and 'best' (NYT), which did not substantively change our findings.

7. We standardize the preregistered Euclidean distance by dividing the pairwise distance between two time series by their length (i.e., the number of months, weeks, days, or hours).

8. Following a reviewer recommendation, we refrain from calculating the preregistered Pearson correlation coefficients for the time series data from Google Trends and instead

extend our preregistered analysis by adding regression analyses based on the Euclidean distance values for Google Trends and the Levenshtein and RBO distance values for the YouTube Data and NYT API.

9. An alternative explanation for the largest differences in samples from the YouTube Data API compared to the other two APIs lies in different regional licensing rights regulating which videos can and cannot be shown in certain countries. However, these licensing rights can only explain differences across download countries but not across consecutive download days.

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APPENDIX

Table A1: Excerpt of a sample downloaded from Google Trends

Search term	Search region	Search time range	Search date	RSI Value
Sofa	DE	2010-01-01 2023-12-31	2010-01-01 00:00:00	25
Sofa	DE	2010-01-01 2023-12-31	2010-02-01 00:00:00	24
Sofa	DE	2010-01-01 2023-12-31	2010-03-01 00:00:00	22
Sofa	DE	2010-01-01 2023-12-31	2010-04-01 00:00:00	19
Sofa	DE	2010-01-01 2023-12-31	2010-05-01 00:00:00	21
Sofa	DE	2010-01-01 2023-12-31	2010-06-01 00:00:00	17
Sofa	DE	2010-01-01 2023-12-31	2010-07-01 00:00:00	18
Sofa	DE	2010-01-01 2023-12-31	2010-08-01 00:00:00	22
Sofa	DE	2010-01-01 2023-12-31	2010-09-01 00:00:00	23
Sofa	DE	2010-01-01 2023-12-31	2010-10-01 00:00:00	26
Sofa	DE	2010-01-01 2023-12-31	2010-11-01 00:00:00	27
Sofa	DE	2010-01-01 2023-12-31	2010-12-01 00:00:00	21
Sofa	DE	2010-01-01 2023-12-31	2011-01-01 00:00:00	30
Sofa	DE	2010-01-01 2023-12-31	2011-02-01 00:00:00	27
Sofa	DE	2010-01-01 2023-12-31	2011-03-01 00:00:00	25
Sofa	DE	2010-01-01 2023-12-31	2011-04-01 00:00:00	20
Sofa	DE	2010-01-01 2023-12-31	2011-05-01 00:00:00	22
Sofa	DE	2010-01-01 2023-12-31	2011-06-01 00:00:00	22
Sofa	DE	2010-01-01 2023-12-31	2011-07-01 00:00:00	26
Sofa	DE	2010-01-01 2023-12-31	2011-08-01 00:00:00	28
Sofa	DE	2010-01-01 2023-12-31	2011-09-01 00:00:00	27
Sofa	DE	2010-01-01 2023-12-31	2011-10-01 00:00:00	32
Sofa	DE	2010-01-01 2023-12-31	2011-11-01 00:00:00	31
Sofa	DE	2010-01-01 2023-12-31	2011-12-01 00:00:00	29
Sofa	DE	2010-01-01 2023-12-31	2012-01-01 00:00:00	36

Note: Data downloaded from Google Trends on April 17, 2024, from Australia for the search term ‘Sofa’ googled in Germany between 2010 and 2023.

Table A2: Excerpt of a sample downloaded from the YouTube API's search endpoint

Search term	Upload location	Upload location radius	Uploaded after	Uploaded before	Upload date	Video id	Video rank
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2013-09-17T13:38:02Z	nIb-GZ6iLkM	1
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2010-11-18T13:31:01Z	7LshWf7e4wY	2
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2014-09-29T20:55:32Z	1VNLZcU7aJI	3
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2010-11-18T13:35:27Z	6VAFnf286-w	4
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2015-04-10T14:48:42Z	fYa7wmvKbiA	5
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2014-02-24T23:01:10Z	gHggFeKQYW4	6
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2010-11-18T13:30:45Z	TgD7M45VXj0	7
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2011-12-03T17:52:37Z	kMktrC6wcUo	8
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2014-01-19T18:43:23Z	ZBuNlzzeIGw	9
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2014-10-03T21:22:16Z	xWZvwGWUz-0	10
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2010-11-18T13:45:27Z	nkKJC7R3Tig	11
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2014-10-07T17:14:28Z	lSA-cqVYU6k	12
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2014-08-25T13:11:02Z	OeRML8guElg	13
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2011-02-17T02:07:44Z	s5hxE_k4JFQ	14
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2014-10-07T17:29:52Z	TKZplawXQVI	15
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2015-01-11T18:59:01Z	lcWtiFL904s	16
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2011-11-05T20:46:50Z	_PvVitJDm_o	17
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2014-08-19T18:27:18Z	HTQCrHvyoME	18
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2014-12-30T11:25:45Z	VrqlIbGwtu0	19
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2014-11-10T08:01:08Z	CEAqCJxD2A0	20
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2013-06-21T22:35:40Z	pm8qe50JU94	21
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2015-01-18T06:27:14Z	UuzXt87L5PU	22
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2015-01-06T10:47:18Z	sw_EpKacpW4	23
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2011-09-19T12:58:35Z	5nXjK7IbDRU	24
Fitness	51.1638175,10.4478313	876km	2005-04-23T00:00:00Z	2015-04-23T00:00:00Z	2013-01-04T18:07:17Z	qHr-WIll9ls	25

Note: Data downloaded from the YouTube API on August 13, 2025, from Australia for the search term 'Fitness' uploaded in Germany between 2005 and 2015.

Table A3: Excerpt of a sample downloaded from the New York Times' Article Search API

Search term	Publication location	Published after	Published before	Publication date	Article id	Article rank
politics	de	19960122	20060122	2005-10-10T05:00:00Z	nyt://article/07fa3f04-7dcc-	1
politics	de	19960122	20060122	2005-08-01T05:00:00Z	nyt://article/7947d021-af2a	2
politics	de	19960122	20060122	2005-07-20T05:00:00Z	nyt://article/fcec5fec-e21e-	3
politics	de	19960122	20060122	2005-06-01T05:00:00Z	nyt://article/be597486-8bd	4
politics	de	19960122	20060122	2005-06-28T05:00:00Z	nyt://article/83c31848-de5	5
politics	de	19960122	20060122	2005-08-05T05:00:00Z	nyt://article/0d2d9584-981	6
politics	de	19960122	20060122	2005-05-31T05:00:00Z	nyt://article/48bf8bd8-584	7
politics	de	19960122	20060122	2005-11-16T05:00:00Z	nyt://article/9778ba77-281	8
politics	de	19960122	20060122	2005-11-02T05:00:00Z	nyt://article/15e2962a-3b7	9
politics	de	19960122	20060122	2005-11-22T05:00:00Z	nyt://article/5727544e-75c	10
politics	de	19960122	20060122	2005-11-15T05:00:00Z	nyt://article/3336fbfc-2048	11
politics	de	19960122	20060122	2005-06-27T05:00:00Z	nyt://article/4d707a38-d43	12
politics	de	19960122	20060122	2005-11-01T05:00:00Z	nyt://article/6f6fb039-3433	13
politics	de	19960122	20060122	2005-10-18T05:00:00Z	nyt://article/2d1a2498-d01	14
politics	de	19960122	20060122	2005-10-07T05:00:00Z	nyt://article/19d87586-966	15
politics	de	19960122	20060122	2005-10-31T05:00:00Z	nyt://article/a8617871-fe50	16
politics	de	19960122	20060122	2005-11-23T05:00:00Z	nyt://article/c2c52f1d-c012	17
politics	de	19960122	20060122	2005-04-01T05:00:00Z	nyt://article/54c7b9ed-3cde	18
politics	de	19960122	20060122	2005-10-11T05:00:00Z	nyt://article/59b64e3f-553	19
politics	de	19960122	20060122	2005-09-20T05:00:00Z	nyt://article/cbc6a246-2a7c	20
politics	de	19960122	20060122	2005-10-06T05:00:00Z	nyt://article/c9b684a4-709	21
politics	de	19960122	20060122	2005-09-29T05:00:00Z	nyt://article/75ff83f3-ed07-	22
politics	de	19960122	20060122	2005-09-21T05:00:00Z	nyt://article/308382d1-17b	23
politics	de	19960122	20060122	2005-08-26T05:00:00Z	nyt://article/79b87948-304	24
politics	de	19960122	20060122	2005-09-20T05:00:00Z	nyt://article/dcab7da9-fe88	25

Note: Data downloaded from the NYT's Article Search API on August 13, 2025, from Australia for the search term 'politics' published about Germany between 1996 and 2006.

Table A4: Standardized Euclidean distance showing the distance between samples returned by Google Trends downloaded from different countries on the same day averaged per search term, time range, and region across all download country pairs and download days

Across all queries for	Mean	Median	Min	Max
Sofa	0.2	0.2	0	0.5
Dachdecker	0.2	0.2	0	1.1
Kurzarbeit	0.4	0	0	3.4
2010-2023	0.1	0	0	0.3
Last 5 years	0.1	0	0	0.3
Last 90 days	0.3	0.2	0	0.8
Last 7 days	0.5	0.3	0	3.4
Worldwide	0.2	0.1	0	0.8
Germany	0.3	0.1	0	3.4

Note: Data downloaded from Google Trends between April 18 and 24, 2024, from Germany, Austria, the United States, and Australia. Larger Euclidean distances attest larger dissimilarities in the samples from different download countries.

Table A5: Standardized Euclidean distances showing the distance between samples returned by Google Trends downloaded from different countries on the same day and for the median across download days

Search term	Search region	Search time	Comparison of Euclidean distance across download countries from... ...single download days: Mean	Median	Min	Max	...median of samples from 7 download days: Mean	Median	Min	Max	Comparison of Euclidean distance across download days: Mean	Median	Min	Max
Sofa	World	2010-2023	0.1	0.1	0	0.3	0.1	0.1	0	0.1	0.2	0.2	0.1	0.3
		Last 5 years	0.1	0.1	0	0.2	0.1	0.1	0	0.1	0.2	0.1	0.1	0.2
		Last 90 days	0.2	0.2	0	0.3	0.1	0.1	0	0.1	0.3	0.3	0.2	0.4
		Last 7 days	0.2	0.3	0	0.5	0.1	0.1	0	0.1	0.7	0.5	0.3	2.8
	Germany	2010-2023	0.1	0.1	0	0.2	0	0.1	0	0.1	0.1	0.1	0.1	0.2
		Last 5 years	0.1	0.2	0	0.3	0.1	0.1	0	0.1	0.2	0.2	0.2	0.4
		Last 90 days	0.1	0.1	0	0.2	0.1	0.1	0	0.1	0.2	0.2	0.1	0.2
		Last 7 days	0.2	0.3	0	0.4	0.1	0.1	0	0.1	0.8	0.6	0.3	3.2
Roofer (Dachdecker)	World	2010-2023	0.2	0.2	0	0.2	0.1	0.1	0	0.1	0.2	0.2	0.2	0.4
		Last 5 years	0.1	0.1	0	0.3	0.1	0.1	0	0.1	0.1	0.1	0	0.3
		Last 90 days	0.2	0.2	0	0.4	0.1	0.1	0	0.1	0.3	0.3	0.2	0.5
		Last 7 days	0.2	0.2	0	0.4	0.1	0.1	0	0.1	0.6	0.5	0.2	2.5
	Germany	2010-2023	0.1	0.1	0	0.3	0.1	0.1	0	0.1	0.2	0.2	0.1	0.3
		Last 5 years	0.1	0.1	0	0.2	0	0	0	0	0.1	0.1	0	0.3
		Last 90 days	0.2	0.2	0	0.3	0.1	0.1	0	0.1	0.2	0.2	0.2	0.3
		Last 7 days	0.5	0.6	0	1.1	0.2	0.3	0.1	0.3	1.3	1.1	0.5	4.3
Short-time work (Kurzarbeit)	World	2010-2023	0	0	0	0	0	0	0	0	0	0	0	0
		Last 5 years	0	0	0	0	0	0	0	0	0	0	0	0
		Last 90 days	0.5	0.6	0	0.8	0.2	0.2	0	0.3	0.8	0.8	0.5	1
		Last 7 days	0.3	0.4	0	0.7	0.1	0.2	0.1	0.2	1	0.8	0.4	4.7
	Germany	2010-2023	0	0	0	0	0	0	0	0	0	0	0	0
		Last 5 years	0	0	0	0	0	0	0	0	0	0	0	0
		Last 90 days	0.4	0.6	0	0.7	0.2	0.2	0	0.3	0.7	0.7	0.5	1
		Last 7 days	1.7	1.7	0	3.4	0.8	0.9	0.4	1.1	3.7	2.7	1.1	22.3

Note: Data downloaded from Google Trends between April 18 and 24, 2024, from Germany, Austria, the United States, and Australia. Larger Euclidean distances attest larger dissimilarities in the samples from different download countries.

Table A6: Jaccard distance showing the distance between samples returned by the YouTube Data API downloaded from different countries on the same day averaged per search term, time range, and region across all download country pairs and download days

Across all queries for	Mean	Median	Min	Max
Fitness	0.6	0.7	0	1
Kurzarbeit	0.2	0.2	0	0.7
Bananenmilch	0.3	0.2	0	0.8
2005-2015	0.3	0.2	0	1
Last 5 years	0.5	0.6	0	0.9
Last month	0.4	0.5	0	1
Last week	0.3	0.2	0	1
Worldwide	0.4	0.3	0	1
Germany	0.3	0.2	0	1

Note: Data downloaded from the YouTube Data API's search endpoint between August 11 and 17, 2025, from Germany, Austria, the United States, and Australia. Larger Jaccard distances attest larger dissimilarities in the samples from different download countries.

Table A7: Jaccard distances showing the distance between samples returned by the YouTube Data API downloaded from different countries on the same day and for the union of samples from all download days

Search term	Upload region	Upload time	Comparison of Jaccard distance across download countries from... ...single download days				...median of samples from 7 download days				Comparison of Jaccard distance across download days			
			Mean	Median	Min	Max	Mean	Median	Min	Max	Mean	Median	Min	Max
Fitness	World	2005 - 2015	0.8	0.9	0.1	1	0.7	0.8	0.3	0.9	0.6	0.6	0.1	0.9
		Last 5 years	0.7	0.8	0.3	0.9	0.7	0.8	0.4	0.8	0.6	0.6	0.2	0.9
		Last month	0.7	0.7	0.1	0.9	0.6	0.6	0.4	0.7	0.7	0.7	0.3	0.9
		Last week	0.6	0.6	0.2	0.9	0.5	0.6	0.2	0.7	0.7	0.7	0.3	0.9
	Germany	2005 - 2015	0.5	0.6	0	1	0.6	0.8	0.1	0.8	0.4	0.3	0	0.9
		Last 5 years	0.6	0.7	0.1	0.8	0.6	0.7	0.4	0.7	0.6	0.6	0.2	0.8
		Last month	0.6	0.7	0.2	1	0.6	0.7	0.5	0.7	0.7	0.7	0.2	1
		Last week	0.6	0.6	0	1	0.6	0.6	0.5	0.7	0.7	0.7	0.1	1
Short-time work (Kurzarbeit)	World	2005 - 2015	0.2	0.2	0	0.3	0.2	0.3	0.1	0.3	0.1	0	0	0.4
		Last 5 years	0.4	0.4	0	0.7	0.4	0.4	0.2	0.5	0.2	0.1	0	0.5
		Last month	0.4	0.6	0	0.7	0.4	0.4	0.1	0.6	0.2	0.2	0	0.6
		Last week	0.1	0.1	0	0.1	0	0.1	0	0.1	0.1	0.1	0	0.3
	Germany	2005 - 2015	0	0	0	0	0	0	0	0	0	0	0	0
		Last 5 years	0.3	0.3	0	0.7	0.4	0.5	0.2	0.5	0.3	0.1	0	0.8
		Last month	0.2	0.2	0	0.2	0.2	0.2	0	0.2	0	0	0	0
		Last week	0.2	0.2	0	0.5	0.2	0.2	0	0.2	0	0	0	0.3
Banana milk (Bananen-milch)	World	2005 - 2015	0.2	0.3	0	0.4	0.3	0.4	0	0.4	0	0	0	0.2
		Last 5 years	0.5	0.6	0.1	0.8	0.5	0.5	0.3	0.6	0.2	0.1	0	0.8
		Last month	0.2	0.2	0	0.5	0.1	0.2	0	0.2	0.2	0.2	0	0.5
		Last week	0.1	0.1	0	0.2	0.1	0.1	0	0.2	0.2	0.2	0	0.4
	Germany	2005 - 2015	0	0	0	0	0	0	0	0	0	0	0	0
		Last 5 years	0.4	0.6	0	0.7	0.4	0.5	0.1	0.6	0.2	0.1	0	0.4
		Last month	0.3	0.5	0	0.5	0.3	0.5	0	0.5	0	0	0	0
		Last week	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

Note: Data downloaded from the YouTube Data API's search endpoint between August 11 and 17, 2025, from Germany, Austria, the United States, and Australia. Larger Jaccard distances attest larger dissimilarities in the samples from different download countries.

Table A8: Jaccard distance showing the distance between samples returned by the NYT API downloaded from different countries on the same day averaged per search term, time range, and region across all download country pairs and download days

Across all queries for	Mean	Median	Min	Max
Politics	0	0	0	0
Soccer	0	0	0	0
Sofa	0.1	0	0	0.4
1996 - 2006	0	0	0	0
Last 5 years	0	0	0	0.3
Last month	0	0	0	0.4
Last week	0	0	0	0.3
Worldwide	0	0	0	0.4
Germany	0	0	0	0

Note: Data downloaded from the NYT Article Search API between August 11 and 17, 2025, from Germany, Austria, the United States, and Australia. Larger Jaccard distances attest larger dissimilarities in the samples from different download countries.

Table A9: Jaccard distances showing the distance between samples returned by the NYT API downloaded from different countries on the same day and for the union of samples from all download days

Search term	Publication region	Publication time	Comparison of Jaccard distance across download countries from... ...single download days				...median of samples from 7 download days				Comparison of Jaccard distance across download days			
			Mean	Median	Min	Max	Mean	Median	Min	Max	Mean	Median	Min	Max
Politics	World	1996 - 2006	0	0	0	0	0	0	0	0	0	0	0	0
		Last 5 years	0	0	0	0	0	0	0	0	0	0	0	0
		Last month	0	0	0	0	0	0	0	0	0	0	0	0
		Last week	0	0	0	0	0	0	0	0	0	0	0	0
	Germany	1996 - 2006	0	0	0	0	0	0	0	0	0	0	0	0
		Last 5 years	0	0	0	0	0	0	0	0	0	0	0	0
		Last month	0	0	0	0	0	0	0	0	0	0	0	0
		Last week	0	0	0	0	0	0	0	0	0	0	0	0
Soccer	World	1996 - 2006	0	0	0	0	0	0	0	0	0	0	0	0
		Last 5 years	0	0	0	0	0	0	0	0	0	0	0	0
		Last month	0	0	0	0	0	0	0	0	0	0	0	0
		Last week	0	0	0	0	0	0	0	0	0	0	0	0
	Germany	1996 - 2006	0	0	0	0	0	0	0	0	0	0	0	0
		Last 5 years	0	0	0	0	0	0	0	0	0	0	0	0
		Last month	0	0	0	0	0	0	0	0	0	0	0	0
		Last week	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Sofa	World	1996 - 2006	0	0	0	0	0	0	0	0	0	0	0	0
		Last 5 years	0.2	0.2	0.1	0.3	0	0	0	0.1	0.3	0.3	0.1	0.5
		Last month	0.2	0.2	0	0.4	0.1	0.1	0	0.1	0.3	0.3	0.1	0.5
		Last week	0.1	0.1	0	0.3	0	0	0	0.1	0.2	0.2	0	0.5
	Germany	1996 - 2006	0	0	0	0	0	0	0	0	0	0	0	0
		Last 5 years	0	0	0	0	0	0	0	0	0	0	0	0
		Last month	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
		Last week	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

Note: Data downloaded from the NYT Article Search API between August 11 and 17, 2025, from Germany, Austria, the United States, and Australia. Larger Jaccard distances attest larger dissimilarities in the samples from different download countries.

TABLES AND FIGURES

Table 1: Overview of download and query parameters for the data collection from the Google Trends, YouTube, and New York Times Article Search API

	Download parameters		Query parameters			
API	**Download countries**	**Download days**	**Search terms**		**Search region**	**Search time ranges**
Google Trends	Germany Austria United States Australia	April 18 - 24, 2024	High search volume: Medium: Low:	Sofa Dachdecker (roofer) Kurzarbeit (short-time work)	Germany Worldwide	January 2010 - December 2023[a] Last 5 years[b] Last 90 days[c] Last 7 days[d]
YouTube Data	Germany Austria United States Australia	August 11 - 17, 2025	High video volume: Medium: Low:	Fitness Kurzarbeit (short-time work) Bananenmilch (banana milk)	Germany Worldwide	April 2005 - April 2015 Last 5 years Last month Last week
New York Times	Germany Austria United States Australia	August 11 - 17, 2025	High article volume: Medium: Low:	Politics Soccer Sofa	Germany Worldwide	January 1996 - January 2006 Last 5 years Last month Last week

Note: We adjusted the query parameters to each API's specificities. Google Trends presents available time ranges and time units: [a] monthly data for time ranges with > 5 years; [b] weekly data for the last five years; [c] daily data for the last 90 days; [d] hourly data for the last seven days up to the past hour.

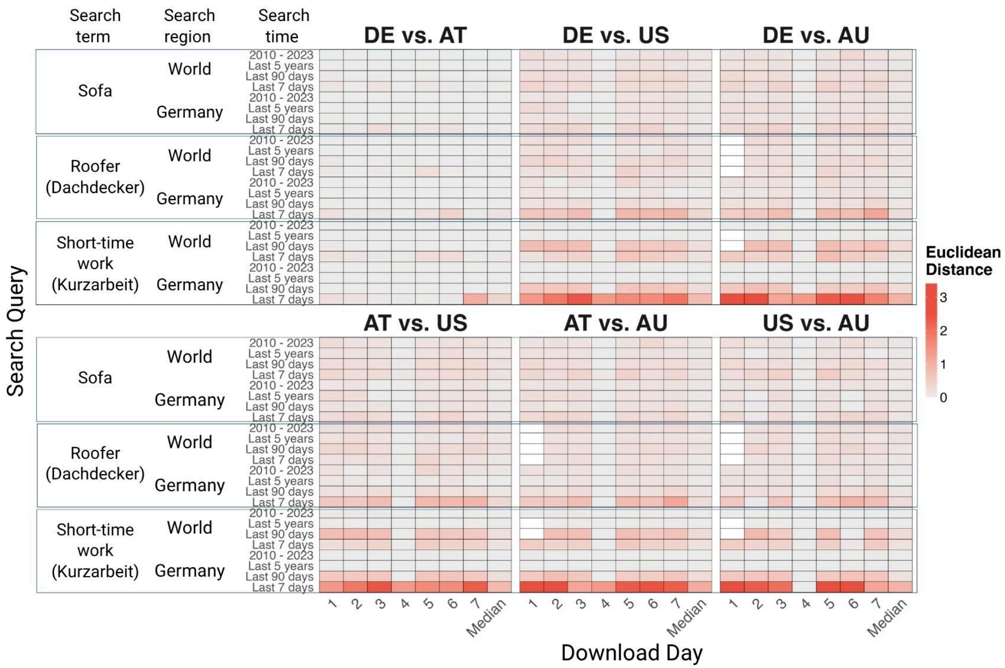

Figure 1: Heatmap of standardized Euclidean distances showing the similarity of Google Trends' time series values between download country pairs across queries per download day and for median across download days

Note: Each cell represents the standardized Euclidean distance between the normalized search interest values from two different download countries downloaded on a given download day and for the median across download days, with darker red shades indicating larger dissimilarities. Columns correspond to country pair comparisons on a given download day and the median across download days, and rows to search queries (combinations of search term, region, and time range). Data were retrieved from Google Trends between April 18 and April 24, 2024, from Germany (DE), Austria (AT), the United States (US), and Australia (AU). Empty cells denote missing values where Google Trends returned no data for the specified query parameters.

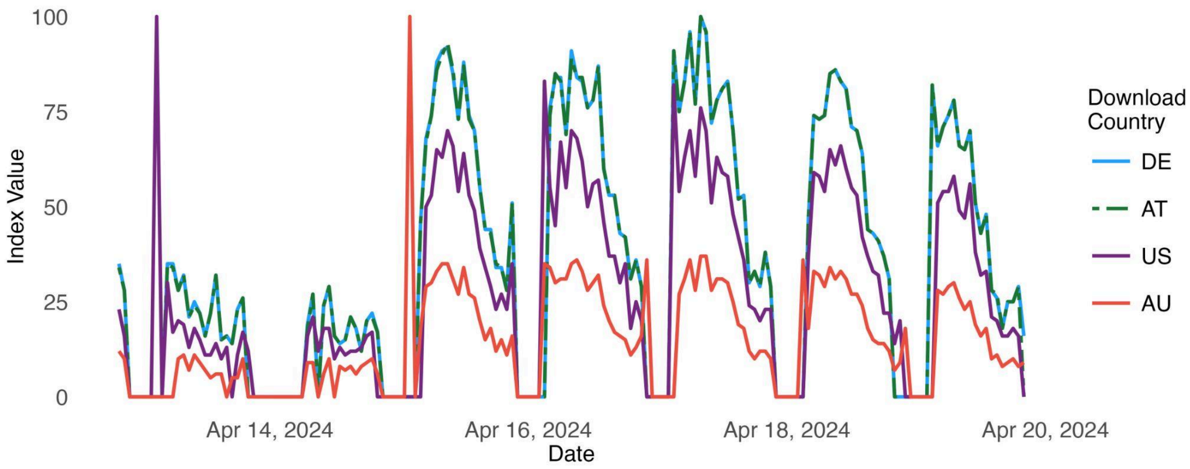

Figure 2: Hourly Google Trend for 'short-time work' ('Kurzarbeit') in Germany from April 15 to April 22, 2024, retrieved on April 19, 2024

Note: Data downloaded from Google Trends on April 19, 2024, from Germany (DE), Austria (AT), the United States (US), and Australia (AU). Trends from Germany and Austria mostly overlap.

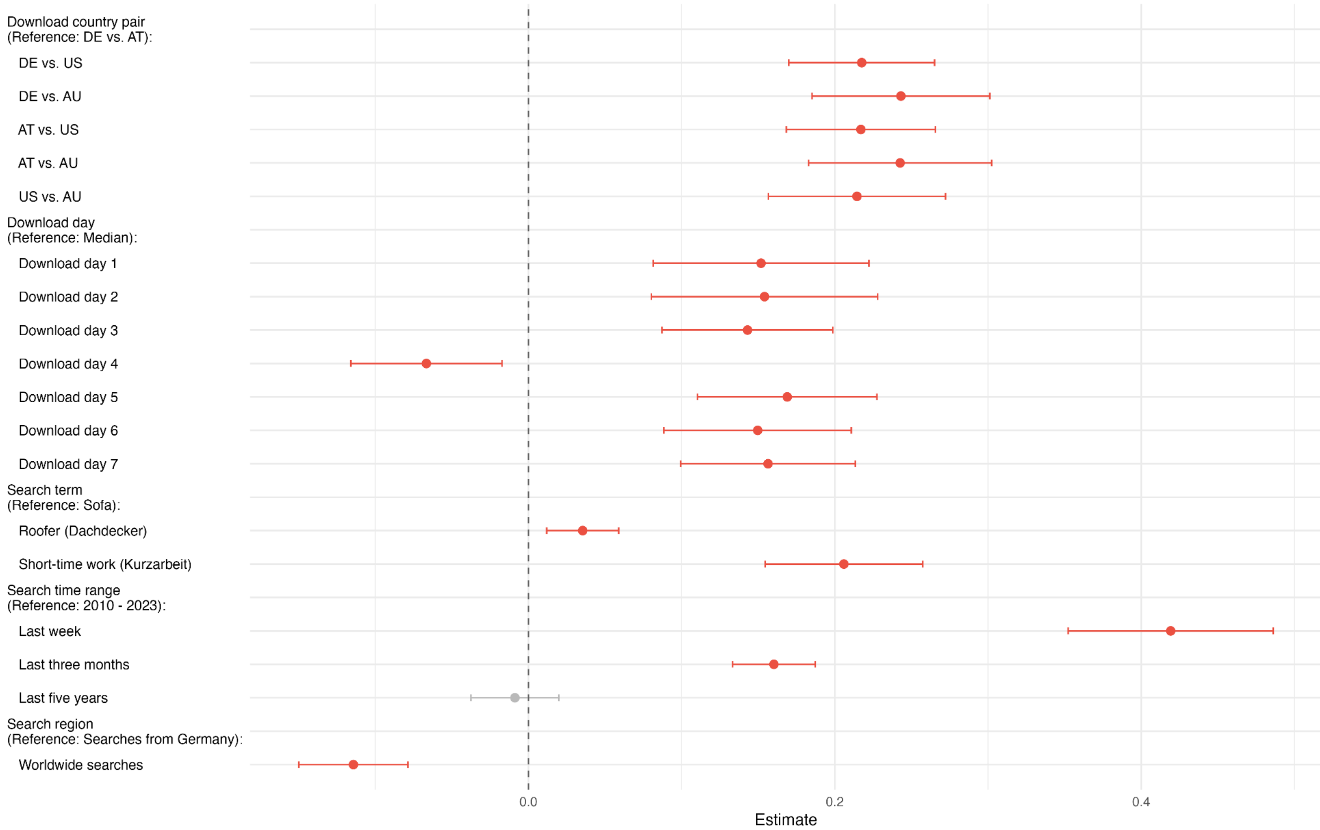

Figure 3: Regression coefficients for the standardized Euclidean Distance between Google Trends time series downloaded from different countries (x-axis), regressed on the download country pair, download day, and query parameters (y-axis)

Note: Reported coefficients are estimated using OLS with robust standard errors. Coefficients statistically significant at the 0.05 level are highlighted in red. Data downloaded from Google Trends between April 18 and 24, 2024, from Germany (DE), Austria (AT), the United States (US), and Australia (AU). The Euclidean distance is standardized by dividing the pairwise distance between two time series by their length (i.e., the number of months, weeks, days, or hours).

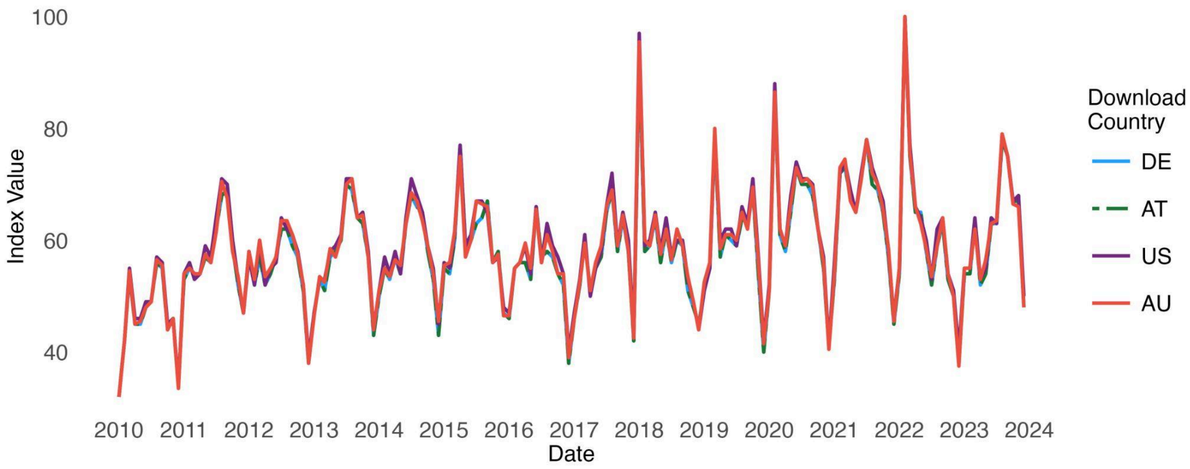

Figure 4: Monthly Google Trend for ‘roofer’ (‘Dachdecker’) in Germany from January 2010 to December 2023 from median of samples from 7 download days

Note: Data downloaded from Google Trends between April 18 and 24, 2024, from Germany (DE), Austria (AT), the United States (US), and Australia (AU). Trends from download countries mostly overlap.

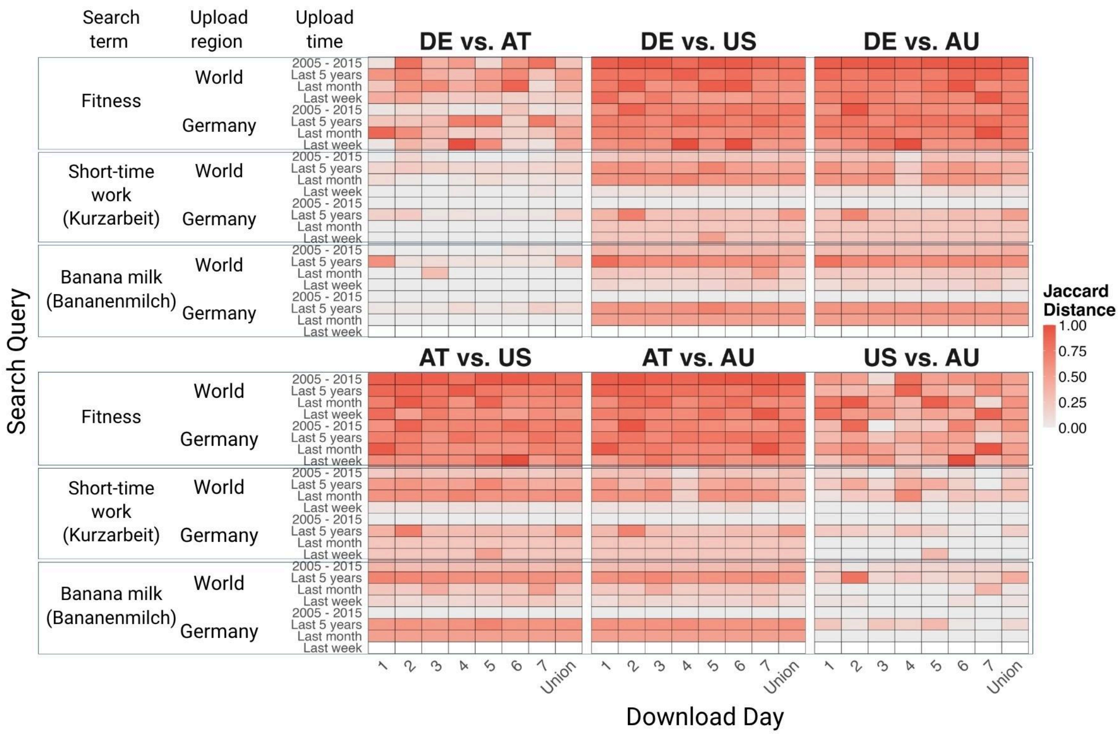

Figure 5: Heatmap of Jaccard distances showing the similarity of YouTube videos between download country pairs across queries per download day and for median across download days

Note: Each cell represents the Jaccard distance between the video sets from two different download countries downloaded on a given download day and for the median across download days, with darker red shades indicating larger dissimilarities. Columns correspond to country pair comparisons on a given download day and the median across download days, and rows correspond to search queries (combinations of search term, region, and time range). Data were retrieved from the YouTube Data API's search endpoint between August 11 and 17, 2025, from Germany (DE), Austria (AT), the United States (US), and Australia (AU). Empty cells denote missing values where the YouTube Data API returned no data for the specified query parameters.

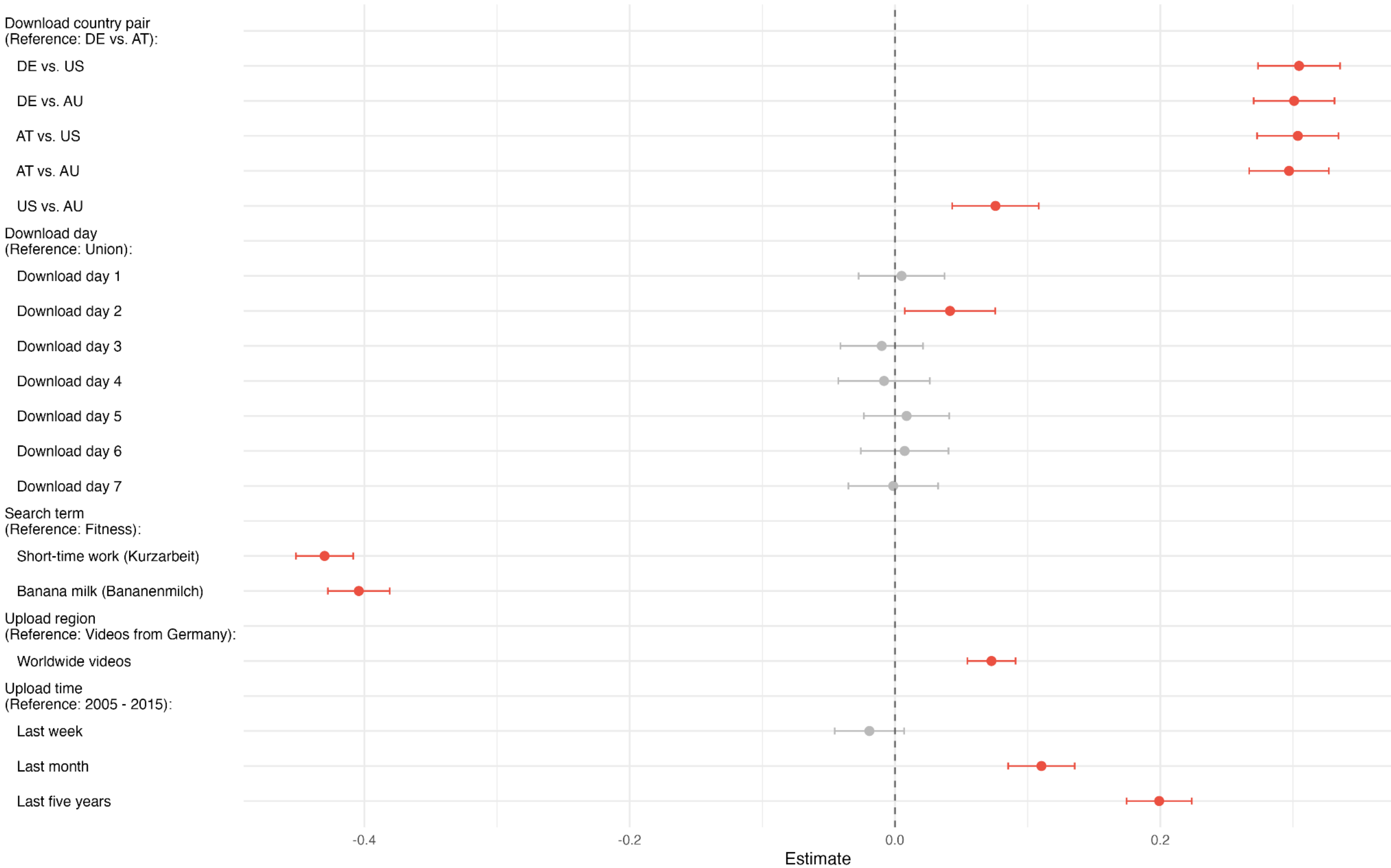

Figure 6: Regression coefficients for the Jaccard Distance between YouTube video sets downloaded from different countries (x-axis), regressed on the download country pair, download day, and query parameters (y-axis)

Note: Reported coefficients are estimated using OLS with robust standard errors. Coefficients statistically significant at the 0.05 level are highlighted in red. Data were retrieved from the YouTube Data API's search endpoint between August 11 and 17, 2025, from Germany (DE), Austria (AT), the United States (US), and Australia (AU).

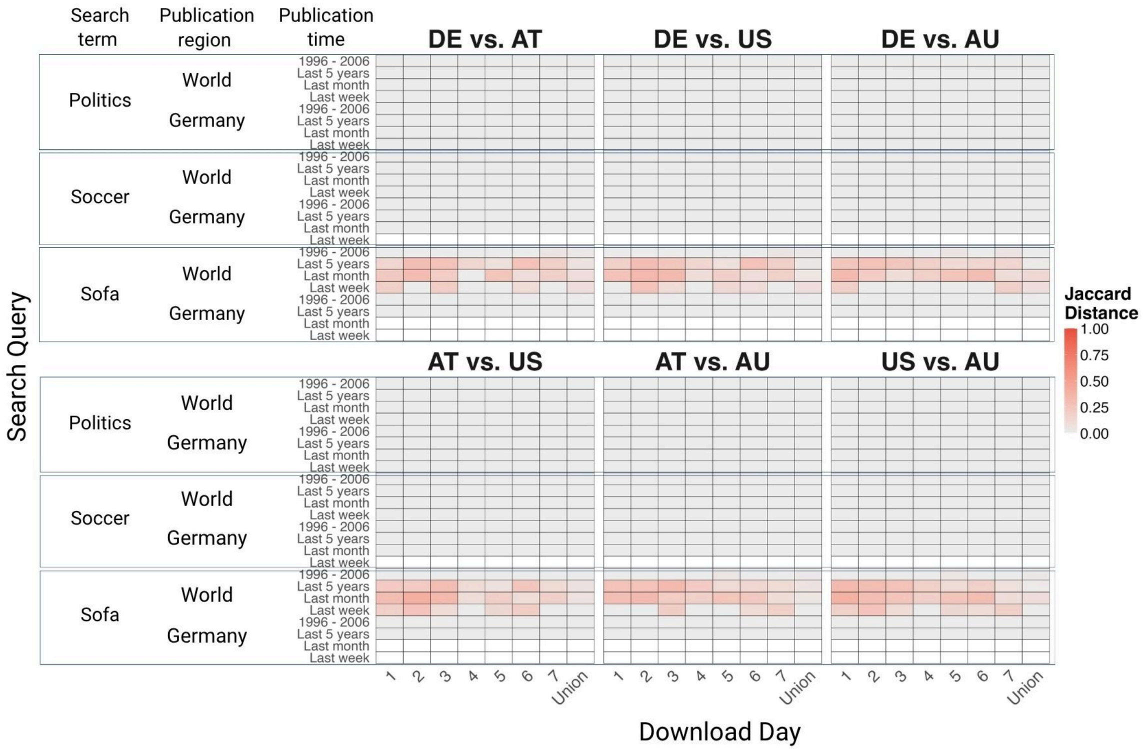

Figure 7: Heatmap of Jaccard distances showing the similarity of New York Times articles between download country pairs across queries per download day and for median across download days

Note: Each cell represents the Jaccard distance between the article sets from two different download countries downloaded on a given download day and for the median across download days, with darker red shades indicating larger dissimilarities. Columns correspond to country pair comparisons on a given download day and the median across download days, and rows correspond to search queries (combinations of search term, region, and time range). Data were retrieved from the New York Times Search API between August 11 and 17, 2025, from Germany (DE), Austria (AT), the United States (US), and Australia (AU). Empty cells denote missing values where the New York Times Search API returned no data for the specified query parameters.

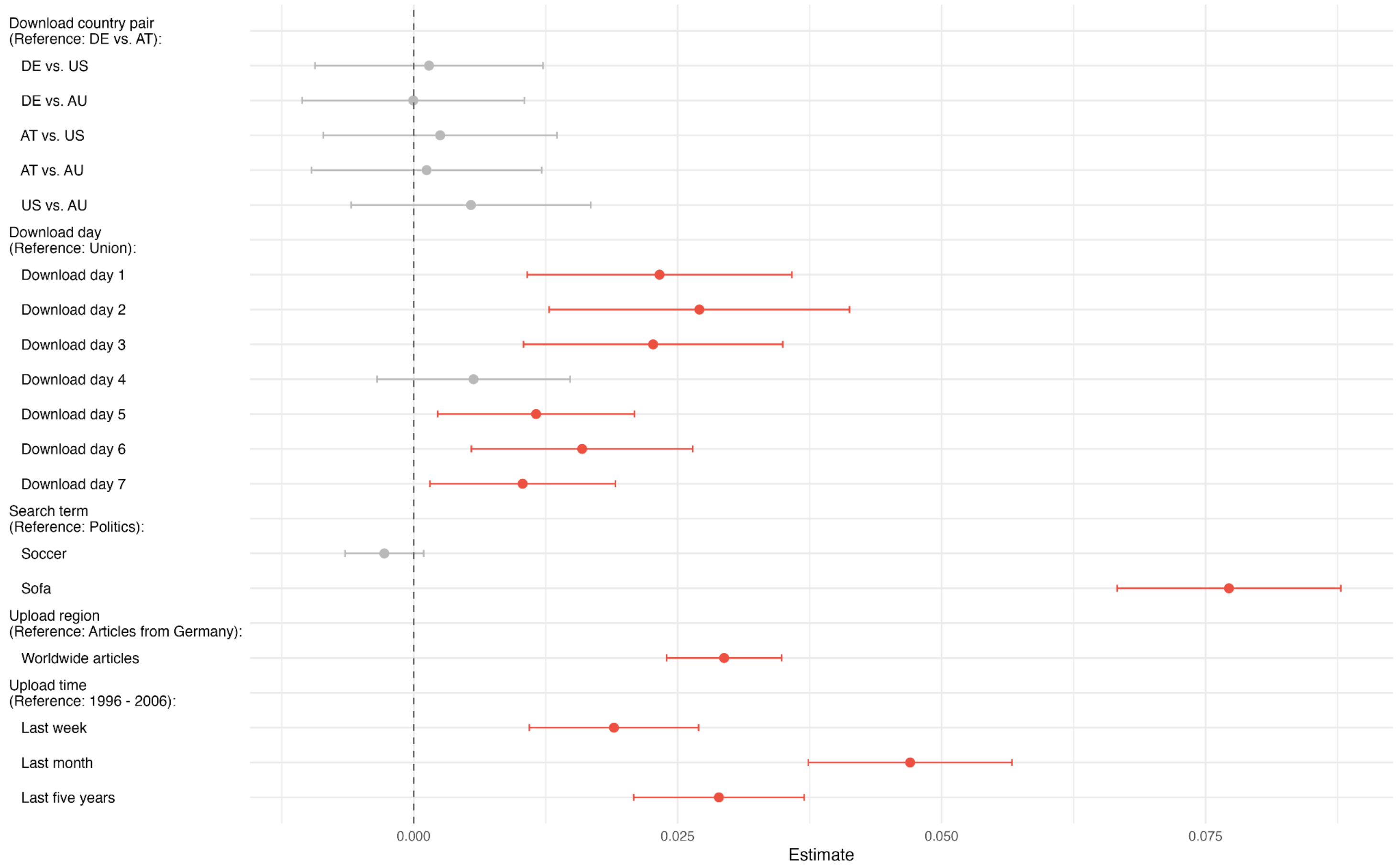

Figure 8: Regression coefficients for the Jaccard Distance between New York Times article sets downloaded from different countries (x-axis), regressed on the download country pair, download day, and query parameters (y-axis)

Note: Reported coefficients are estimated using OLS with robust standard errors. Coefficients statistically significant at the 0.05 level are highlighted in red. Data were retrieved from the New York Times Search API between August 11 and 17, 2025, from Germany (DE), Austria (AT), the United States (US), and Australia (AU).